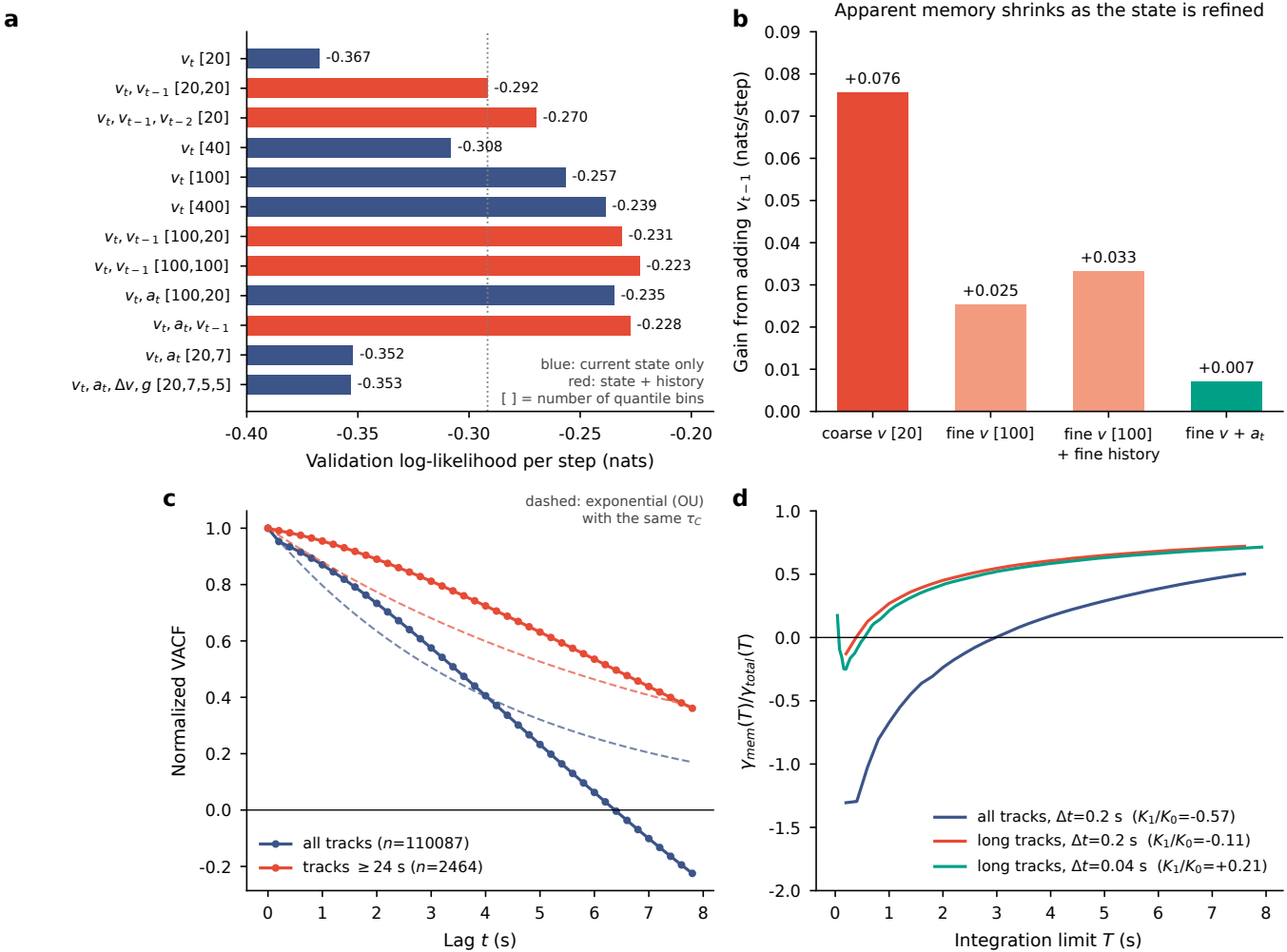
a
v_t [20]
v_t, v_{t-1} [20,20]
v_t, v_{t-1}, v_{t-2} [20]
v_t [40]
v_t [100]
v_t [400]
v_t, v_{t-1} [100,20]
v_t, v_{t-1} [100,100]
v_t, a_t [100,20]
v_t, a_t, v_{t-1}
v_t, a_t [20,7]
$v_t, a_t, \Delta v, g$ [20,7,5,5]
-0.367
-0.292
-0.270
-0.308
-0.257
-0.239
-0.231
-0.223
-0.235
-0.228
-0.352
-0.353
blue: current state only
red: state + history
[ ] = number of quantile bins
-0.40
-0.35
-0.30
-0.25
-0.20
Validation log-likelihood per step (nats)
b
Apparent memory shrinks as the state is refined
Gain from adding v_{t-1} (nats/step)
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
+0.076
+0.025
+0.033
+0.007
coarse v [20]
fine v [100]
fine v [100] + fine history
fine $v + a_t$
c
dashed: exponential (OU) with the same τ_c
Normalized VACF
1.0
0.8
0.6
0.4
0.2
0.0
-0.2
all tracks (n=110087)
tracks ≥ 24 s (n=2464)
0
1
2
3
4
5
6
7
8
Lag t (s)
d
$\gamma_{mem}(T)/\gamma_{total}(T)$
1.0
0.5
0.0
-0.5
-1.0
-1.5
-2.0
all tracks, Δt=0.2 s (K_1/K_0=-0.57)
long tracks, Δt=0.2 s (K_1/K_0=-0.11)
long tracks, Δt=0.04 s (K_1/K_0=+0.21)
0
1
2
3
4
5
6
7
8
Integration limit T (s)